Bangladesh University of Professionals (BUP)
Department of Information and Communication Technology (ICT)
Program: B.Sc. in ICE, BICE-2024
Mid Term Examination
Course Title: Digital Communication
Course Code: ICE-3105

Date: 09.04.26 **Full Marks: 30**
Time: 1.00 Hour

Instructions:

A. **Answering Question-1 is mandatory for achieving CO1. You must answer Question-1 first, then the others.**
B. **Answer any two (02) questions from 2 to 4.**
C. **All parts of a question must be answered together.**
D. **Marks are shown in the margin.**
E. **Symbols used carry their usual meaning.**

Question-1 CO 1

a. A sensor in an industrial automation system produces a continuous voltage signal X. The Cumulative Distribution Function (CDF) of the signal is modeled by the following logistic growth function. 4

$$CDF_X(x) = \frac{1}{1+e^{-x}}$$

Formula – 1.5
Math – 2.5

Evaluate the probability that the amplitude of the signal at any given time will be in between 1 and 5 Volts inclusive. Show your calculation steps.

b. A stochastic signal $X(t)$ represents the output of a high-frequency sensor. The statistical behavior of the signal amplitude is described by the following Probability Density Function (PDF). 3

$$PDF_X(x) = \begin{cases} 5e^{-x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$$

correct formula, otherwise 1
1.5+1.5

Calculate the total average power of the signal.

c. **Demonstrate** whether the PDF given in Question-1(b) satisfies the Exponential Distribution or not. 3

μ 1
σ 1
Condition 1

Question-2

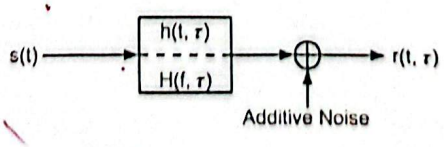
receiver
Analog → reconstruct/
Digital → identify/ recognize

a. Explain the advantage of a Digital Communication System over an Analog Communication System from the receiver's perspective. 2

b. Describe the responsibilities of a transmitter in a Digital Communication System in detail. 4

per responsibility 0.5 / Explanation → 1

c. Discuss the properties of the filter used in the following channel. Explain the distortions that the channel introduces. 4



identify → 1
per info. → 0.5
per distort → 0.5

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Question-3

a. At any given time, the transmitter of a Digital Communication System receives the following string from its information source. 4

W W W X # X # # # K K K L L W W K % L W

Table -2 | Tree -2 | incomplete -1

Find out the Huffman codes for the symbols. Ignore the spaces in between. wrong combination -0.5

b. Define Information and Entropy from the perspective of Digital Communication. Calculate the Information and Entropy from the Huffman codes found in Question-3(a). 3

0.5 0.5 1 0.5+0.5 1 0.5+0.5

same for all { I(x) H(x)

c. Explain efficiency of a source coding scheme. Calculate the efficiency for the Huffman coding found in Question-3(a) 3

1 ratio

may vary { L - 0.5+0.5 η - 0.5+0.5

unit not considered

Question-4

a. Explain the causes of inter-symbol interference. 2

1 cause → 1
2 cause → 2 / No Explanation -0.5

b. Discuss the encoding and decoding procedures of a Linear Predictive Coding model. 5

per equation → 1
per other points → 0.5

c. Describe the characteristics and applications of a Baseband Digital Transmission System. 3

per point → 0.5

0 Hz → ∞ Hz → 1

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